

Activity 1: Network Protocol Analysis with Wireshark

Objective:

Learn how to capture and analyze packets to understand TCP/IP, addressing, and protocol behavior.

Steps:

1. **Install Wireshark** (free: <https://www.wireshark.org>).
2. Open Wireshark → Select active network interface (e.g., Wi-Fi).
3. Start a packet capture and browse a few websites.
4. Stop the capture after ~1 min.
5. Apply filters:
 - ip → Shows only IP packets.
 - tcp → Shows only TCP packets.
 - http → Shows HTTP traffic.
6. Analyze packet headers:
 - **Source/Destination IPs**
 - **Protocols used (TCP, UDP, ICMP)**
 - **Port numbers**
 - **Round Trip Time (RTT)**
7. Save the capture as evidence for your lab report.

Activity 2: RF Propagation Modeling with Python

Objective:

Simulate **Free-Space Path Loss (FSPL)** to understand wireless coverage.

Formula:

$$\text{FSPL(dB)} = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.44$$
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Where:

- d = distance (km)
- f = frequency (MHz)

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
frequency = 2400 # MHz (2.4 GHz WiFi)
distances = np.linspace(0.1, 20, 100) # km

# FSPL calculation
fspl = 20*np.log10(distances) + 20*np.log10(frequency) + 32.44

# Plot
plt.figure(figsize=(8,5))
plt.plot(distances, fspl, label="FSPL at 2.4 GHz")
plt.xlabel("Distance (km)")
plt.ylabel("Path Loss (dB)")
plt.title("Free-Space Path Loss vs Distance")
plt.grid(True)
```

```
plt.legend()
```

```
plt.show()
```